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39. Broken staybolts

a) Maximum allowable number of broken staybolts
No boiler shall be allowed to remain in service with two broken staybolts located within 24 inches of each other, as measured inside the firebox or combustion chamber on a straight line. No boiler shall be allowed to remain in service with more than 4 broken staybolts inside the entire firebox and combustion chamber, combined

b) Staybolt replacement
Broken staybolts must be replaced during the 31 service day inspection, if detected at that time. Broken staybolts detected in between 31 service day inspections must be replaced no later than 30 calendar days from the time of detection. When staybolts 8 inches or less in length are replaced, they shall be replaced with bolts that have telltale holes 3/16 inch to 7/32 inch in diameter and at least 1 1/4 inches deep at each end, or that have telltale holes 3/16 inch to 7/32 inch in diameter their entire length. At the time of replacement of broken staybolts, adjacent staybolts shall be inspected.

c) Assessment of broken staybolts
Telltale holes leaking, plugged, or missing shall be counted as broken staybolts.

d) Prohibited methods of closing telltale holes.
Welding, forging, or riveting broken staybolt ends is prohibited as a method of closing telltale holes.

40. Time and method of staybolt testing

a) Time of hammer testing
   i. General.
   All staybolts shall be hammer tested at every 31 service day inspection, except as provided in paragraph (a)(ii) of this Section. All staybolts also shall be hammer tested under hydrostatic pressure any time hydrostatic pressure above the MAWP specified on the boiler specification Form No. 4 is applied to the boiler. (See appendix B.)

   ii. Exception for inaccessible staybolts.
   The removal of brickwork or grate bearers for the purpose of hammer testing staybolts during each 31 service day inspection will not be required if the staybolts behind these structural impediments have a telltale hole 3/16 inch to 7/32 inch in diameter their entire length. Whenever the brickwork or grate bearers are removed for any other reason, however, the bolts shall be inspected at that time.